

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Lecture 15: Final Project

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Overview

The final project combines one new observable with the existing fit and documents the resulting parameter shift.

1 Physical assumptions

All coordinate conditions and boundary conditions are stated before a parameter is fitted.

$$Registry_{final} = O_1, \dots, O_{52}, O_*$$

2 Field equations

The first two displayed relations establish the notation used in the derivation.

$$L_{53} < L_{accept}$$

3 Observable limit

Weak-field, static, and point-source limits are checked independently.

$$hash(\theta_{final}, log, registry) = submission_id$$

4 Parameter dependence

Parameter combinations are compared through derivatives of the reference observables.

5 Example

The calculation is repeated for one analytic case and one tabulated data point.

Checks

- Verify dimensions and sign conventions.
- State the boundary conditions used in each limit.
- Report numerical precision separately from model uncertainty.

Reading

- Final Closure Protocol r2
- SGT9 branch and merge guide
- Research authorship standard MAS-54